



American International University Bangladesh

Computer Science & Engineering

Faculty Name : Juena Ahmed Noshin

Course Name : Introduction To Database

Section: K

Project Name: Bus Ticket Management System

Name	ID	Contribution
Shihab Khan	22-49326-3	70%
Reduanul Islam	22-49323-3	30%

✦ CONTENTS:	PAGE NO
✦ Introduction -----	2
✦ Scenario Description -----	3
✦ ER Diagram-----	4
✦ Normalization-----	5-10
✦ Schema Diagram-----	11
✦ Table Creation-----	12-15
✦ Data Insertion -----	16-19
✦ Query Writin-----	20-28
✦ Relational Algebra -----	29
✦ Conclution-----	30

✚ INTRODUCTION:

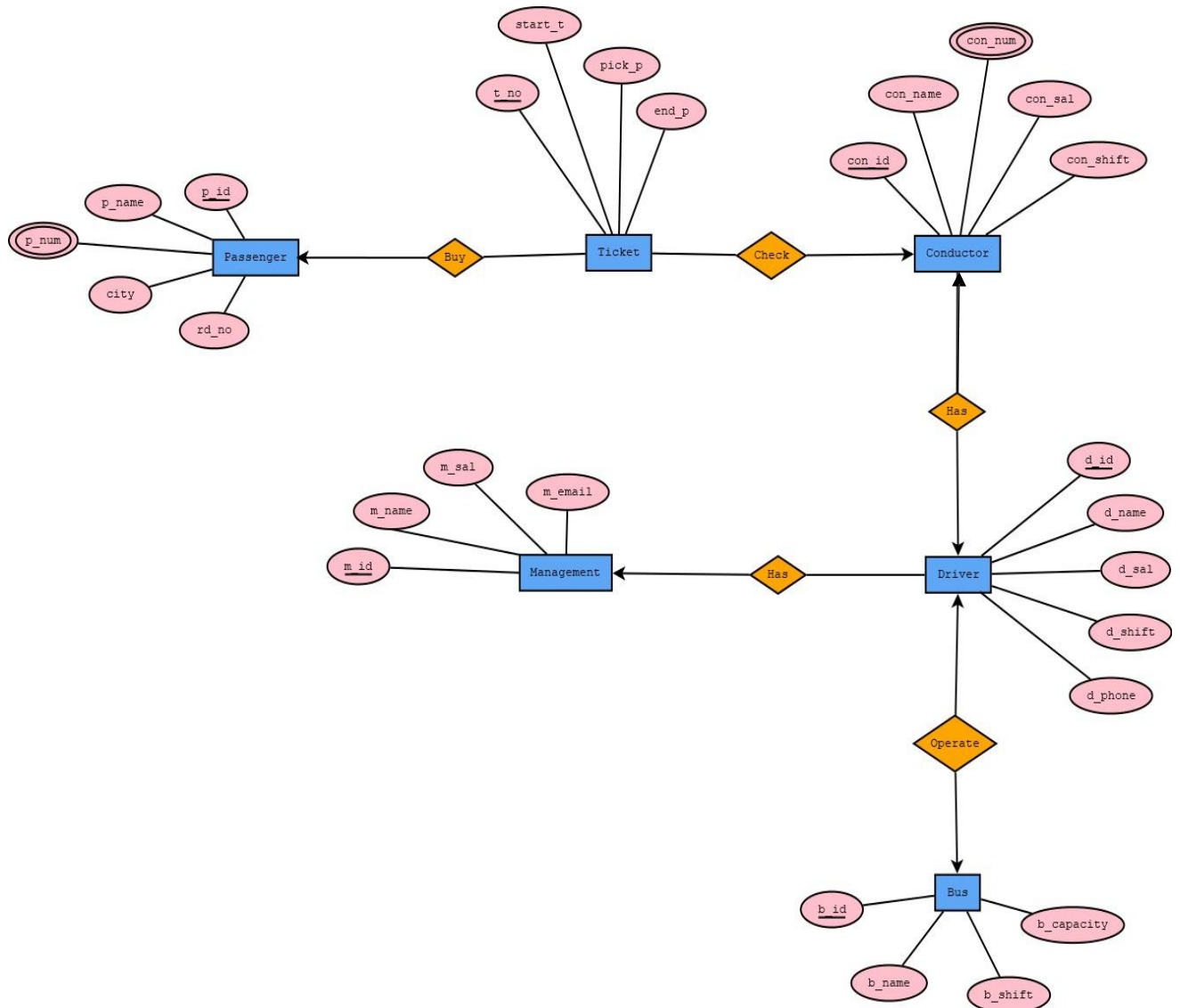
This is a project about a bus ticket management system. It introduces an advanced platform that effortlessly weaves together employee, passenger, ticket management, driver and other data. The following system helps us to find our specific data. It's like a helpful guide for exploring and utilizing data.

The purpose of this project is to design and implement a project that stores and manages information about employees, passengers, drivers, conductor, route, bus and other relevant data. The system is designed to support a small-scale bus ticket management system with a potential for scalability in the future.

† SCENARIO DESCRIPTION :

In a bus management system a passenger can buy many tickets. One ticket can be bought by one passenger. System also stores passengers p_id, p_name, p_number, city, rd_no. A passenger address is composed of city, rd_no. A ticket has t_no, start_t, pick_p, end_t. There is a management. Management stores m_id, m_name, m_sal, m_email. Management has many Drivers. Each Driver has d_id, d_name, d_sal, d_shift, d_phone. A driver has a conductor. A conductor is identified by con_id, con_name, con_num, con_sal, con_shift. A Conductor can check many tickets. A bus stores b_id, b_name, b_shift, b_capacity. One bus is operated by one driver.

† ER DIAGRAM:



Normalization:

Buy:

UNF:

Buy(p_id,p_name,p_num,city,rd_no,t_no,t_price,end_p,
start_t,pick_p)

1NF:

p_num is multivalued attribute.

1. p_id,p_name,p_num,city,rd_no,t_price,t_no,end_p,
start_t,pick_p

2NF:

1. p_id,p_name,p_num,city,rd_no
2. t_no,t_price,end_p,start_t,pick_p

3NF:

1. p_id,p_name,p_num
2. city,rd_no
3. t_no, t_price,end_p,start_t,pick_p

Table creation:

1. p_id,p_name,p_num,**a_id**.
2. t_no,t_price,end_p,start_t,pick_p,**p_id**
3. a_id,city,rd_no.

Check:

UNF:

Check(t_no,t_price,end_p,start_t,pick_p,
con_id,con_name,con_num,con_sal,con_shift)

1NF:

con_num is multivalued attribute.

1. t_no,t_price,end_p,start_t,pick_p,
con_id,con_name,con_num,con_sal,con_shift

2NF:

1. t_no,t_price,end_p,start_t,pick_p,
2. con_id,con_name,con_num,con_sal,con_shift

3NF:

There is no transitive dependency .

1. t_no,t_price,end_p,start_t,pick_p,
2. con_id,con_name,con_num,con_sal,con_shift

Table creation:

- 1.t_no,t_price,end_p,start_t,pick_p,con_id
- 2.con_id,con_name,con_num,con_sal,con_shift.

Has

UNF

Has(con_id,con_name,con_sal,con_num,con_shift,d_id,d_name,d_sal,d_shift,
d_phone)

1NF

con_num is multivalued attribute.

1. con_id,con_name,con_sal,con_num,con_shift,d_id,d_name,d_sal,d_shift,
d_phone

2NF

1. con_id,con_name,con_sal,con_num,con_shift
2. d_id,d_name,d_sal,d_shift,d_phone

3NF:

There is no transitive dependency .

1. con_id,con_name,con_sal,con_num,con_shift
3. d_id,d_name,d_sal,d_shift,d_phone

Table creation:

1. con_id,con_name,con_sal,con_num,con_shift,**d_id**
2. d_id,d_name,d_sal,d_shift,d_phone

Operate:

UNF

Operate(d_id,d_name,d_sal,d_shift,d_phone,b_id,b_name,b_shift,b_capacity)

1NF

There is no multivalued attribute.

1. d_id,d_name,d_sal,d_shift,d_phone,b_id,b_name,b_shift,b_capacity

2NF

1. d_id,d_name,d_sal,d_shift,d_phone
2. b_id,b_name,b_shift,b_capacity

3NF:

There is no transitive dependency .

1. d_id,d_name,d_sal,d_shift,d_phone
2. b_id,b_name,b_shift,b_capacity

Table creation:

1. d_id,d_name,d_sal,d_shift,d_phone
2. b_id,b_name,b_shift,b_capacity,d_id.

Has:

UNF

Operate(d_id,d_name,d_sal,d_sift,d_phone,m_id,m_name,m_sal,m_email)

1NF

There is no multivalued attribute.

1. d_id,d_name,d_sal,d_sift,d_phone,m_id,m_name,m_sal,m_email

2NF

1. d_id,d_name,d_sal,d_sift,d_phone
2. m_id,m_name,m_sal,m_email

3NF:

There is no transitive dependency .

1. d_id,d_sal,d_name,d_sift,d_phone
2. m_id,m_name,m_sal,m_email

Table creation:

1. d_id,d_sal,d_name,d_sift,d_phone,**m_id**
2. m_id,m_name,m_sal,m_email.

Temporary table:

1. p_id,p_name,p_num,**a_id**.
2. t_no,t_price,end_p,start_t,pick_p,**p_id**,
3. a_id,city,rd_no.
4. t_no,t_price,end_p,start_t,pick_p,**con_id**
5. ~~con_id,con_name,con_num,con_sal,con_shift.~~
6. con_id,con_name,con_sal,con_num,con_shift,**d_id**
7. ~~d_id,d_name,d_sal,d_shift,d_phone~~
8. ~~d_id,d_name,d_sal,d_shift,d_phone~~
9. b_id,b_name,b_shift,b_capacity,**d_id**.
10. d_id,d_sal,d_name,d_sift,d_phone,**m_id**
11. m_id,m_name,m_sal,m_email

Final table:

1. p_id,p_name,p_num1,p_num2,**a_id**.
2. t_no,t_price,end_p,start_t,pick_p,**p_id**,
3. a_id,city,rd_no.
4. t_no,t_price,end_p,start_t,pick_p,**con_id**
5. con_id,con_name,con_sal,con_num1,con_num2,con_shift,**d_id**
6. b_id,b_name,b_shift,b_capacity,**d_id**.
7. d_id,d_sal,d_name,d_sift,d_phone,**m_id**
8. m_id,m_name,m_sal,m_email

Schemadiagram:

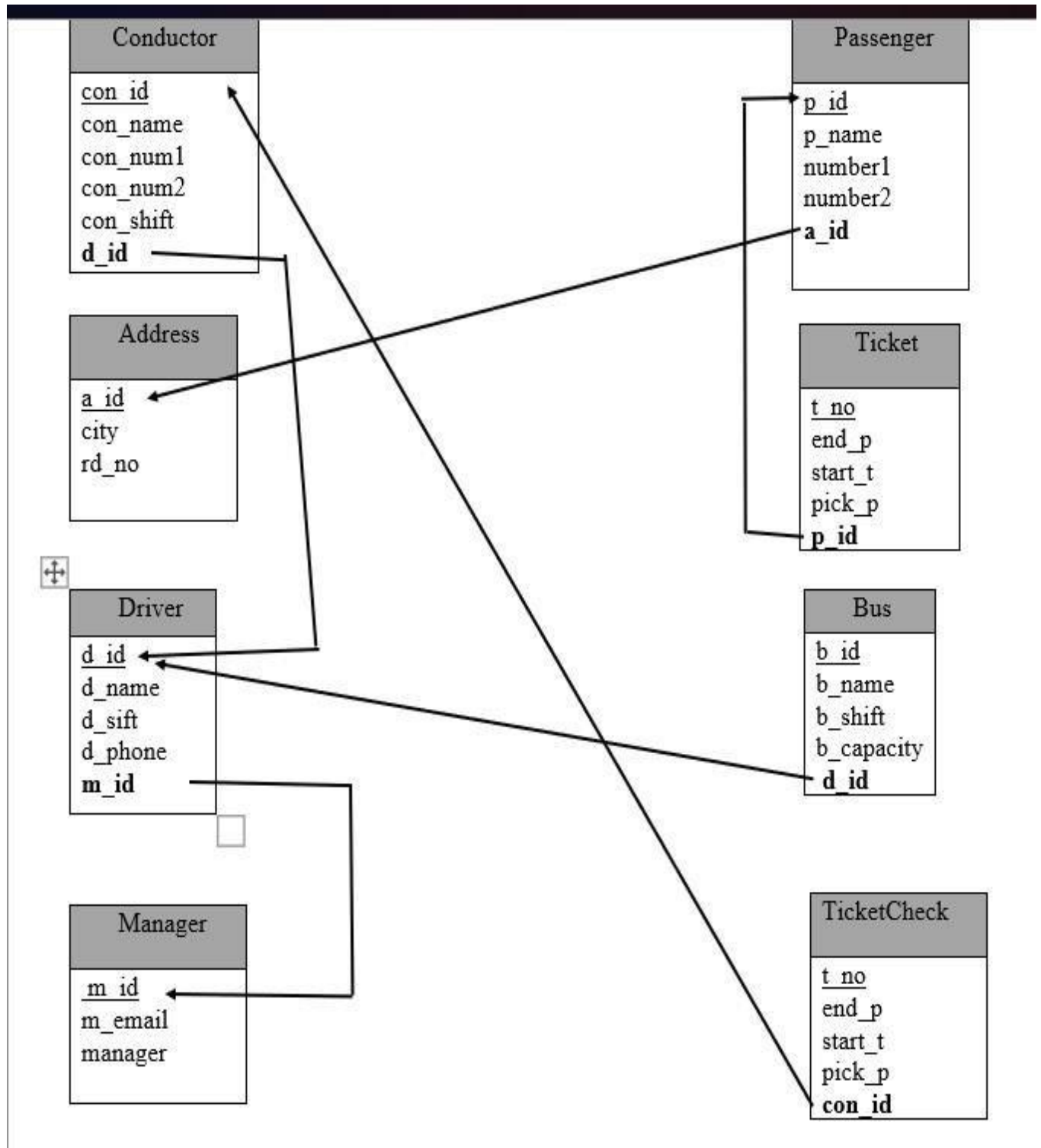


Table Creation

1. Passenger :

☒ Autocommit Display 10 Save Run

```
1.
Create Table Passenger(p_id number primary key,p_name Varchar(200),p_num1 number,p_num2
number,a_id varchar(200));
ALTER TABLE Passenger ADD CONSTRAINT Address_ID FOREIGN KEY (a_id) REFERENCES Address(a_id);
create sequence p_seq increment by 1 start with 1 maxvalue 100 nocache nocycle;
describe Passenger;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **PASSENGER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PASSENGER	P_ID	Number	-	-	-	1	-	-	-
	P_NAME	Varchar2	200	-	-	-	✓	-	-
	P_NUM1	Number	-	-	-	-	✓	-	-
	P_NUM2	Number	-	-	-	-	✓	-	-
	A_ID	Varchar2	200	-	-	-	✓	-	-

1 - 5

2. Ticket :

```
2.
Create Table Ticket (t_no number primary key,
start_t Date,pick_p varchar(200),end_p varchar(200),t_price number,
p_id number);
ALTER TABLE Ticket ADD CONSTRAINT Passenger_ID FOREIGN KEY (p_id) REFERENCES Passenger(p_id);
describe Ticket;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **TICKET**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TICKET	T_NO	Number	-	-	-	1	-	-	-
	START_T	Date	7	-	-	-	✓	-	-
	PICK_P	Varchar2	200	-	-	-	✓	-	-
	END_P	Varchar2	200	-	-	-	✓	-	-
	T_PRICE	Number	-	-	-	-	✓	-	-
	P_ID	Number	-	-	-	-	✓	-	-

1 - 6

5. Conductor :

5.

```
Create Table Conductor(con_id number primary key , con_name varchar(200),con_sal number,con_num2
number,con_num1 number,con_shift varchar(200),d_id number);
alter table Conductor add constraint dd_id foreign key (d_id) references Driver(d_id);
ALTER TABLE Conductor ADD CONSTRAINT chk_A_Salary CHECK (con_sal<= 20000 and con_sal>=10000 );
describe conductor;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **CONDUCTOR**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CONDUCTOR	CON_ID	Number	-	-	-	1	-	-	-
	CON_NAME	Varchar2	200	-	-	-	✓	-	-
	CON_SAL	Number	-	-	-	-	✓	-	-
	CON_NUM2	Number	-	-	-	-	✓	-	-
	CON_NUM1	Number	-	-	-	-	✓	-	-
	CON_SHIFT	Varchar2	200	-	-	-	✓	-	-
	D_ID	Number	-	-	-	-	✓	-	-

1 - 7

6. Bus :

6.

```
Create Table Bus(b_id number primary key,b_name varchar(200),b_shift varchar(200),b_capacity
number,d_id number);
ALTER TABLE Bus ADD CONSTRAINT ddd_ID FOREIGN KEY (d_id) REFERENCES Driver(d_id);
create sequence b_seq increment by 1 start with 301 maxvalue 400 nocache nocycle;
describe Bus;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **BUS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
BUS	B_ID	Number	-	-	-	1	-	-	-
	B_NAME	Varchar2	200	-	-	-	✓	-	-
	B_SHIFT	Varchar2	200	-	-	-	✓	-	-
	B_CAPACITY	Number	-	-	-	-	✓	-	-
	D_ID	Number	-	-	-	-	✓	-	-

1 - 5

7. Driver :

☒ Autocommit Display 10 **Save** **Run**

7.
 Create Table Driver(d_id number primary key, d_sal number ,d_name varchar(200),d_sift varchar(200),d_phone number,m_id number);
 Alter table Driver add constraint manager_id foreign key (m_id) references manager(m_id);
 ALTER TABLE Driver ADD CONSTRAINT chk_d_Salary CHECK (d_sal<= 30000 and d_sal>=20000);
 describe Driver;

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **DRIVER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DRIVER	D_ID	Number	-	-	-	1	-	-	-
	D_SAL	Number	-	-	-	-	✓	-	-
	D_NAME	Varchar2	200	-	-	-	✓	-	-
	D_SIFT	Varchar2	200	-	-	-	✓	-	-
	D_PHONE	Number	-	-	-	-	✓	-	-
	M_ID	Number	-	-	-	-	✓	-	-
1 - 6									

8. Manager :

☒ Autocommit Display 10 **Save** **Run**

8.
 Create Table Management (m_id number primary key,m_name Varchar(200),m_sal number ,m_email varchar(200));
 ALTER TABLE Management ADD CONSTRAINT chk_M_Salary CHECK (m_sal<= 50000 and m_sal>=30000);
 describe Management;

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **MANAGEMENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MANAGEMENT	M_ID	Number	-	-	-	1	-	-	-
	M_NAME	Varchar2	200	-	-	-	✓	-	-
	M_SAL	Number	-	-	-	-	✓	-	-
	M_EMAIL	Varchar2	200	-	-	-	✓	-	-
1 - 4									

Data Insertion:

1. Passenger :

P_ID	P_NAME	P_NUM1	P_NUM2	A_ID
1	Mithila	1854516453	1354516452	D_N
2	Faysal	1854516453	1454516452	M_C
3	Rishan	1354516452	1554516453	C_C
4	Eva	1854516454	1854516451	C_K
5	Tushi	1554516453	1854516452	C_S
6	Mahi	1354516454	1354516452	D_N
7	Fahim	1454516454	185416453	M_C
8	Shihab	1854516453	1854516453	C_C
9	Redu	1854516453	1854516453	C_K
10	Rakib	1854516453	1854516453	C_S

2.Ticket :

T_NO	START_T	PICK_P	END_P	T_PRICE	P_ID
111	01-JAN-23	Dhaka	Mymensingh	500	1
101	02-JAN-23	Mymensingh	Dhaka	500	2
102	03-JAN-23	Chittagong	Cumilla	600	3
103	04-JAN-23	Cumilla	Chittagong	600	4
104	05-JAN-23	Chadpur	Dhaka	400	5
105	06-JAN-23	Dhaka	Chadpur	400	6
106	07-JAN-23	Mymensingh	Cumilla	550	7
107	08-JAN-23	Cumilla	Mymensingh	550	8
108	09-JAN-23	Chittagong	Dhaka	450	9
109	10-JAN-23	Dhaka	Chittagong	450	10

3.Address :

A_ID	CITY	RD_NO
D_N	Dhaka	Nikunja/2
M_C	Mymensingh	College road/3
C_C	Chittagong	Chakbazar/3
C_K	Cumilla	Kandirpar/5
C_S	Chandpur	Sharsti/6

5 rows returned in 0.00 seconds

4. TicketCheck :

T_NO	T_PRICE	END_P	START_T	PICK_P	CON_ID
111	500	Mymensingh	01-JAN-23	Dhaka	221
101	500	Dhaka	02-JAN-23	Mymensingh	221
102	600	Cumilla	03-JAN-23	Chittagong	222
103	600	Chittagong	04-JAN-23	Cumilla	222
104	400	Dhaka	05-JAN-23	Chadpur	223
105	400	Chadpur	06-JAN-23	Dhaka	223
106	550	Cumilla	07-JAN-23	Mymensingh	224
107	550	Mymensingh	08-JAN-23	Cumilla	224
108	450	Dhaka	09-JAN-23	Chittagong	225
109	450	Chittagong	10-JAN-23	Dhaka	225

5. Conductor :

CON_ID	CON_NAME	CON_SAL	CON_NUM2	CON_NUM1	CON_SHIFT	D_ID
221	Golam Rabbi	15000	1197349086	1187349087	Day	701
222	Shahin	17000	1197349084	1199349082	Night	702
223	Roni	14000	1197349011	1199349033	Day	703
224	Kholil	18000	1197349073	1199349043	Night	704
225	Aziz	15000	1197349564	1199349856	Day	705

6. Bus :

B_ID	B_NAME	B_SHIFT	B_CAPACITY	D_ID
301	Inception	Day	16	701
302	Inception	Night	16	702
303	Inception	Day	16	703
304	Inception	Night	16	704
305	Inception	Day	16	705

7. Diver :

D_ID	D_SAL	D_NAME	D_SIFT	D_PHONE	M_ID
701	23000	Soud	Day	1123676512	1001
702	27000	Rifat	Night	1123676512	1002
703	24000	Manna	Day	1123676513	1003
704	29000	Shakib	Night	1123676514	1001
705	21000	Maruf	Day	1123676515	1002

8. Management :

M_ID	M_NAME	M_SAL	M_EMAIL
1003	Faysal Khan	49999	faysal07@gmail.com
1001	Shihab Khan	49999	shihab07@gmail.com
1002	Fahim Rafe	49999	fahimrafe07@gmail.com

Query Writing :

1. Single-Row Functions:

Question (a) : Display the names of passengers along with the length of their names.

```
SELECT p_name, LENGTH(p_name)
AS name_length FROM Passenger;
```

Results Explain Describe Saved S

P_NAME	NAME_LENGTH
Mithila	7
Faysal	6
Rishan	6
Eva	3
Tushi	5
Mahi	4
Fahim	5
Shihab	6
Redu	4
Rakib	5

Question (b) : In the "Passenger" table, write a query to display the names of passengers in uppercase.

```
SELECT p_name, UPPER(p_name) as  
uppername_name from passenger;
```

Results Explain Describe Saved S

P_NAME	UPPERNAME_NAME
Mithila	MITHILA
Faysal	FAYSAL
Rishan	RISHAN
Eva	EVA
Tushi	TUSHI
Mahi	MAHI
Fahim	FAHIM
Shihab	SHIHAB
Redu	REDU
Rakib	RAKIB

2. Group Function :

Question(a) : Find the average ticket price.

Home > SQL > **SQL Commands**

☒ Autocommit Display 10 Save Run

```
SELECT AVG(t_price) AS average_ticket_price FROM Ticket;
```

Results Explain Describe Saved SQL History

AVERAGE_TICKET_PRICE
500

1 rows returned in 0.00 seconds [CSV Export](#)

Question(b) : Count the number night shift conductor.

Home > SQL > **SQL Commands**

☒ Autocommit Display 10 Save Run

```
select count(d_sift) from driver group by d_sift;
```

Results Explain Describe Saved SQL History

COUNT(D_SIFT)
2
1
2

3 rows returned in 0.00 seconds [CSV Export](#)

3. Subquery :

Question(a) : Display the details of passengers who have tickets.

Home > SQL > **SQL Commands**

☒ Autocommit Display 10 **Save** **Run**

```
SELECT *FROM Passenger WHERE p_id IN (SELECT DISTINCT p_id FROM Ticket);
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	P_NUM1	P_NUM2	A_ID
1	Mithila	1854516453	1354516452	D_N
2	Faysal	1854516453	1454516452	M_C
3	Rishan	1354516452	1554516453	C_C
4	Eva	1854516454	1854516451	C_K
5	Tushi	1554516453	1854516452	C_S
6	Mahi	1354516454	1354516452	D_N
7	Fahim	1454516454	185416453	M_C
8	Shihab	1854516453	1854516453	C_C
9	Redu	1854516453	1854516453	C_K
10	Rakib	1854516453	1854516453	C_S

10 rows returned in 0.00 seconds

[CSV Export](#)

Question(b) : Find the passengers with a salary greater than the average salary.

Home > SQL > **SQL Commands**

☒ Autocommit Display 10 

Save

Run

```
SELECT * FROM Passenger WHERE p_num1 > (SELECT AVG(p_num1) FROM Passenger);|
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	P_NUM1	P_NUM2	A_ID
1	Mithila	1854516453	1354516452	D_N
2	Faysal	1854516453	1454516452	M_C
4	Eva	1854516454	1854516451	C_K
8	Shihab	1854516453	1854516453	C_C
9	Redu	1854516453	1854516453	C_K
10	Rakib	1854516453	1854516453	C_S

6 rows returned in 0.00 seconds

[CSV Export](#)

4. Joining :

Question(a) : Display the details of passengers who live in Chittagong

```
1->SELECT *FROM Passenger e, Address d where e.A_ID=d.A_ID and d.CITY='Chittagong';|
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	P_NUM1	P_NUM2	A_ID	A_ID	CITY	RD_NO
3	Rishan	1354516452	1554516453	C_C	C_C	Chittagong	Chakbazar/3
8	Shihab	1854516453	1854516453	C_C	C_C	Chittagong	Chakbazar/3

2 rows returned in 0.00 seconds

[CSV Export](#)

Question(b) : Display the passenger name and there city whose id is less then 5

[Home](#) > [SQL](#) > [SQL Commands](#)

☒ Autocommit **Display** 10 **Save** **Run**

```
SELECT e.p_name,d.city FROM Passenger e, Address d where e.a_id=
d.a_id and e.p_id<5 ;
```

Results [Explain](#) [Describe](#) [Saved SQL](#) [History](#)

P_NAME	CITY
Mithila	Dhaka
Faysal	Mymensingh
Rishan	Chittagong
Eva	Cumilla

4 rows returned in 0.00 seconds [CSV Export](#)

5. View :

Question(a) : .Create a view called conductorView based on the con_name and con_sal from the Conductor table.

5.Views:

a.Create a view called conductorView based on the con_name and con_sal from the Conductor table.

```
CREATE VIEW conductorView AS SELECT con_name,  
con_sal FROM conductor;  
select * from conductorView;
```

Results Explain Describe Saved SQL History

CON_NAME	CON_SAL
Golam Rabbi	15000
Shahin	17000
Roni	14000
Kholil	18000
Aziz	15000

Question(b) : Create a view called passenger based on the p_id,p_name,a_id from the passenger table.

```
create view passengerview as select p_id,p_name  
from passenger;  
select * from passengerview;
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	A_ID
1	Mithila	D_N
2	Faysal	M_C
3	Rishan	C_C
4	Eva	C_K
5	Tushi	C_S
6	Mahi	D_N
7	Fahim	M_C
8	Shihab	C_C
9	Redu	C_K
10	Rakib	C_S

Relational Algebra :

1. select those rows of data where pick_p is Dhaka from the Ticket table.

Ans: $\sigma_{\text{pick_p} = \text{"Dhaka"}}(\text{Ticket})$

2. Project the t_no and the t_price from the Ticket table.

Ans: $\Pi_{\text{t_no}, \text{t_price}}(\text{Ticket})$

3. Find the name of Passenger whose p_id is 4

Ans: $\Pi_{\text{p_name}}(\sigma_{\text{p_id} = 4}(\text{Passenger}))$

4. Find the name of the Driver whose shift is Night.

Ans: $\Pi_{\text{d_name}}(\sigma_{\text{d_shift} = \text{"Night"}}(\text{Driver}))$

5. Find the name of conductor whose salary is greater than 15000 taka.

Ans: $\Pi_{\text{c_name}}(\sigma_{\text{con_sal} > 15000}(\text{Conductor}))$

Conclusion :

In conclusion, bus ticket management system will provide a centralized system for sorting and managing information about passengers, tickets, conductors, bus time, bus number, managements, bus details, bus drivers. The system will allow users to generate reports, search and filter information, and create, update, delete records as necessary. All kinds of normalizations were done properly so no data will be lost while working with this management system. This project will be implemented using Oracle 10G. In summary, by creating a unified repository for sorting and managing a wealth of critical information, including details about bus, passengers, drivers, tickets, conductor, bus time, bus number, managements, bus details, bus drivers this system addresses the multifaceted needs of modern bus ticket management systems.

